Group 2 – Process Mining Project Roadmap

GitHub repository: github.com/tnen128/Group2_PMProject26

Overview

Group 2 (Mohamed and Abdurrahman) works on dataset 2, LaDe: we turn its raw parcel data into an event log and analyse the pickup process with process mining.

1. **Preprocessing:** build the event log from the raw data, export it as XES and fill in the 37-attribute metadata table.
2. **Process discovery:** discover BPMN models that show how the process is executed.
3. **Process enhancement (Option 2):** find bottlenecks and which attributes, such as courier, area or city, relate to late pickups.

We present at the mid-term on 2 Dec and the final on 28 Jan, and submit a report of max 15 pages (LNCS format) by 11 Feb.

Dataset: LaDe

- Real last-mile parcel data from Cainiao (Alibaba), May to October 2022.
- About 10.7M parcels handled by 21k couriers in five Chinese cities: Shanghai, Hangzhou, Chongqing, Jilin and Yantai.
- Two subsets, one CSV per city: pickup (LaDe-P, 6.1M rows) and delivery (LaDe-D, 4.5M rows).
- Each row is one parcel with its courier, region and area, the time the task was accepted, the time it was picked up or delivered, and GPS points.
- Pickups also have a promised time window, so we can compare plan and reality.
- We start with Jilin, the smallest city (262k pickups), and scale up once the pipeline works.

Source: LaDe on Hugging Face

Tools

- **Python** with pandas for data preparation and PM4Py for the event log, XES export and process discovery
- **Disco** for quick visual checks of the process maps
- **GitHub** repository Group2_PMProject26 for code and notebooks
- **LaTeX** with the LNCS template for the report

Timeline

Event log done by the mid-term in week 5; analysis and writing follow

Project phases by week, with the person who leads each phase

